

Software Architecture and Design Specification

Project: Hospital Management System (HMS)

Version: 1.0

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Status: Draft

Revision History

Version	Date	Author	Change Summary
1.0	12-09-2026	TEAM 13	Initial SAD document based on HMS SRS and Software Test Plan.

Approvals

Role	Name	Signature/Date
Course Coordinator	Dhijo Kumar Behera	
Project Team	TEAM 13	

1. Introduction

1.1 Purpose

This document specifies the proposed software architecture and detailed design of the Hospital Management System (HMS). It translates the functional, non-functional, security, interface, and testing requirements defined in the HMS Software Requirements Specification (SRS) into architectural components, responsibilities, interactions, interfaces, and design decisions.

1.2 Scope

The architecture covers centralized management of hospital operations including user authentication, patient registration and profiles, doctor management, appointment scheduling, medical records, electronic prescriptions, medicine dispensing records, billing and payments, reports, and notifications.

The architecture also addresses role-based access, protection of patient and medical information, secure communication, data integrity, performance, reliability, availability, usability, and maintainability. Internal

implementation of external payment gateways, laboratory systems, insurance systems, and third-party services is outside the project scope.

1.3 Audience

Developers, QA engineers, doctors, nurses and hospital staff, receptionists, pharmacists, administrators, system administrators, project evaluators, and course instructors.

1.4 Definitions

Term	Definition
HMS	Hospital Management System
SRS	Software Requirements Specification
STP	Software Test Plan
RTM	Requirements Traceability Matrix
RBAC	Role-Based Access Control
EMR	Electronic Medical Record
UI	User Interface
API	Application Programming Interface
UAT	User Acceptance Testing
TLS	Transport Layer Security
HTTPS	Hypertext Transfer Protocol Secure

2. Document Overview

2.1 How to Use This Document

This document describes the HMS architecture and design decisions. It should be read together with the SRS, Software Test Plan (STP), UML use-case diagrams, and Requirements Traceability Matrix (RTM). The architecture sections identify major components and their responsibilities; the design sections describe component interactions, interfaces, security controls, error handling, and operational considerations.

2.2 Related Documents

Document	Purpose
Hospital Management System SRS, Version 1.0	Defines functional, non-functional, security, interface, and acceptance requirements.
Hospital Management System Software Test Plan	Defines testing scope, strategy, test levels, test data, schedule, roles, risks, and traceability.
Requirements Traceability Matrix (RTM)	Maps requirements to corresponding test cases and provides requirement coverage.
HMS UML Use-Case Diagrams	Describe major user interactions and system use cases.

3. Architecture

3.1 Goals & Constraints

Area	Architecture Goal / Constraint
Security	Protect patient, medical, prescription, billing, and user information; enforce RBAC; use secure communication; maintain audit logs.
Performance	At least 90% of transactions should complete within 3 seconds under normal operating load.
Reliability & Availability	Maintain consistent operation and target at least 99.5% monthly availability excluding scheduled maintenance.
Data Integrity	Prevent loss or corruption of patient, appointment, medical, prescription, and billing records.
Usability	Provide role-specific dashboards and a clear, consistent

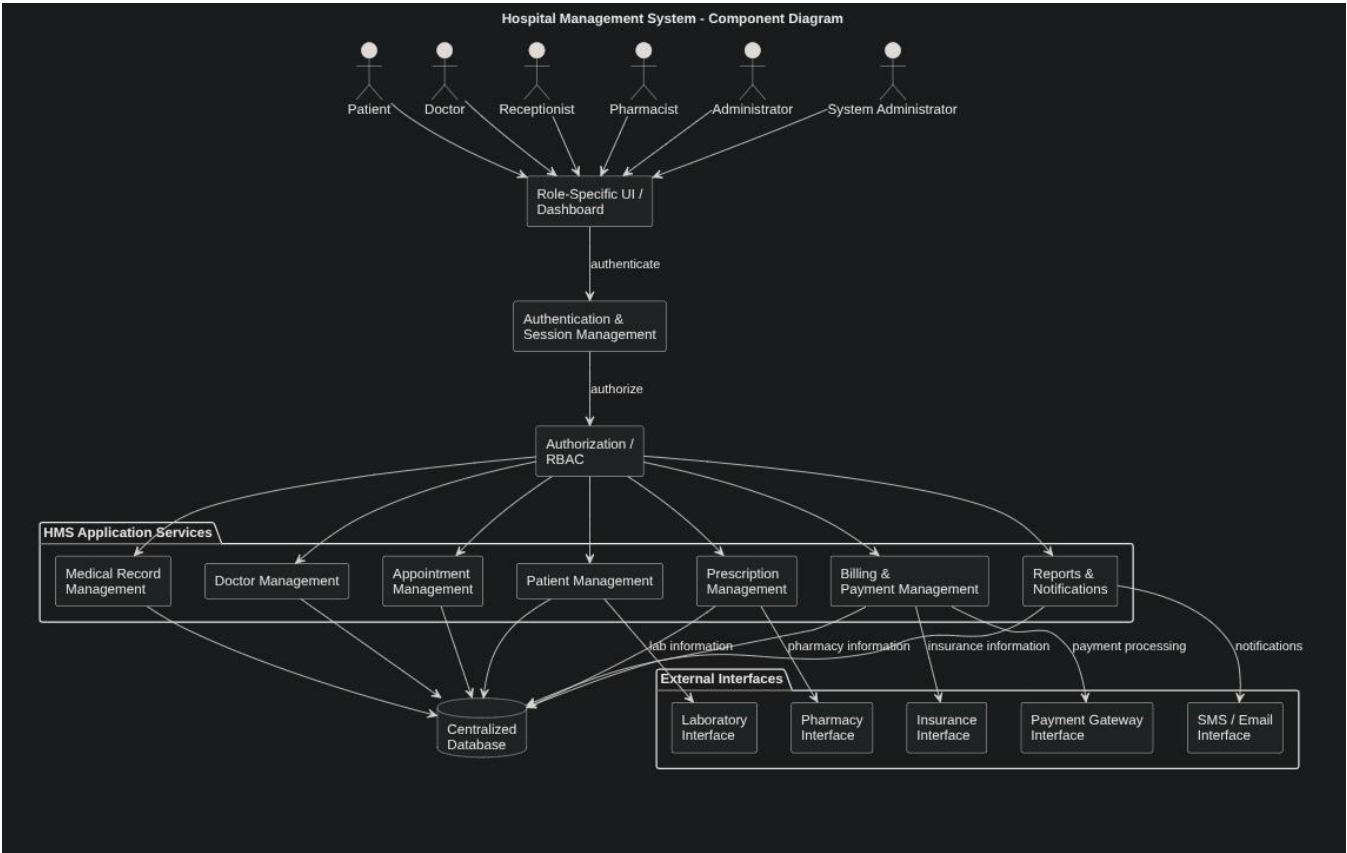
	interface requiring minimal training.
Environment	Support hospital computers, laptops, tablets, modern web browsers or supported application interfaces, a centralized database server, and a secure hospital network/Internet connection.
Security Constraint	Sensitive information must be transmitted using HTTPS with TLS 1.2 or higher.
External Dependencies	Payment, SMS/email, laboratory, pharmacy, and insurance integrations may depend on external APIs.

3.2 Stakeholders & Concerns

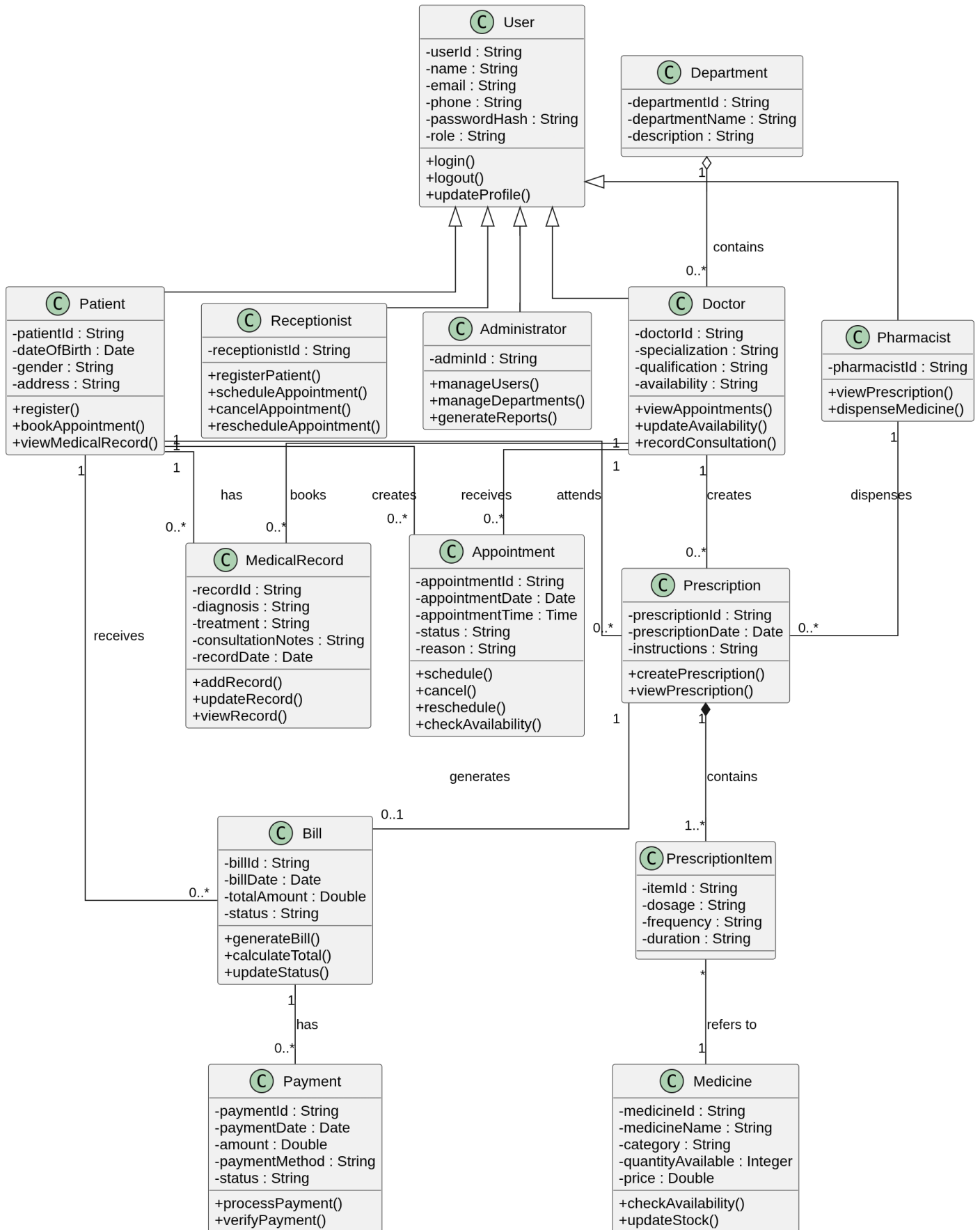
Stakeholder	Primary Concerns
Patient	Secure access, privacy, appointments, medical records, prescriptions, bills, and notifications.
Doctor	Appointments, authorized patient records, diagnosis/treatment recording, and e-prescriptions.
Receptionist	Patient registration, appointments, patient information, and billing.
Pharmacist	Prescription access and medicine dispensing records.
Administrator	User, doctor, department, configuration, and report management.
System Administrator	Maintenance, security, backups, monitoring, and technical administration.
QA / Project Evaluator	Requirement coverage, correctness, security, performance, usability, and acceptance.

3.3 Component and Class Diagram

The following logical component view is derived from the HMS SRS. The exact implementation technology is not specified in the source documents.



Hospital Management System - UML Class Diagram



Suggested component relationships: Users → Role-Specific UI/Dashboard → Authentication & RBAC → HMS Application Services → Domain Modules (Patient, Doctor, Appointment, Medical Record, Prescription, Billing, Reports/Notifications) → Centralized Database. Application services may connect to external Payment, SMS/Email, Laboratory, Pharmacy, or Insurance interfaces through defined APIs.

3.4 Component Descriptions

Component	Responsibilities
Role-Specific User Interface	Provides dashboards and workflows for patients, doctors, receptionists, pharmacists, administrators, and system administrators; validates user input and displays feedback.
Authentication & Session Component	Authenticates users, manages sessions/logout, applies session timeout, and supports secure password handling.
Authorization / RBAC Component	Restricts functions and data according to assigned roles and permissions.
Patient Management	Registers patients, manages permitted demographic/contact information, and provides authorized patient record access.
Doctor Management	Manages doctor profiles, specialization, department, availability, and consultation slots.
Appointment Management	Schedules, validates, cancels, reschedules, and tracks appointments; prevents double booking and supports notifications.
Medical Record Management	Creates, updates, and retrieves authorized medical records, diagnosis, treatment details, and consultation notes.
Prescription Management	Creates electronic prescriptions and manages medicine, dosage, frequency, duration, instructions, and dispensing records.
Billing & Payment Management	Generates bills, records payments, and updates payment status.
Reports & Notification Component	Generates hospital operational reports and supports appointment confirmation/reminder notifications.
Centralized Database	Stores user, role, patient, doctor, department, appointment, medical, prescription, medicine, billing, payment, and related operational data.
External Integration Interfaces	Provide controlled interfaces to external payment, laboratory, insurance, SMS, email, pharmacy, or other supported services where required.

3.5 Chosen Architecture Pattern and Rationale

Proposed pattern: Layered / modular centralized architecture. This choice follows the SRS description of a centralized system and its clear separation of user interfaces, application functions, centralized data storage, and external interfaces. It also supports maintainability and role-specific access.

Layer / Module	Responsibility
Presentation Layer	Role-specific dashboards, forms, validation messages, and user interaction.
Application / Service Layer	Coordinates use cases such as registration, appointment scheduling, consultation, prescription, billing, and reporting.
Security Layer	Authentication, RBAC, session management, password protection, and audit logging.
Domain / Business Modules	Patient, doctor, appointment, medical record, prescription, billing, reporting, and notification logic.
Data Access Layer	Controlled access to the centralized database and consistency of stored records.
Integration Layer	Interfaces with external payment, laboratory, insurance, SMS/email, or related systems.

The SRS does not specify a microservices architecture or a particular framework; therefore no such implementation is assumed in this SAD.

3.6 Technology Stack & Data Stores

The SRS specifies the deployment environment and security protocol but does not prescribe a programming language, application framework, database product, or cloud platform.

Area	Source-Supported Design
Client	Hospital computers, laptops, tablets, or other supported devices using a modern web browser or supported application interface.
Application	HMS application/service layer; exact language and framework: To be finalized.
Database	Centralized database server; exact DBMS: To be finalized.
Communication	HTTPS with TLS 1.2 or higher for sensitive information.
External Services	Payment gateway, SMS/email, laboratory, pharmacy, and insurance interfaces where required.
Logging / Monitoring	Audit logging for security-sensitive actions; system monitoring and operational monitoring to be defined during implementation.

3.7 Risks & Mitigations

Risk	Mitigation / Design Response
Centralized database unavailable	Maintain suitable availability, backup, recovery, and monitoring procedures.
Network connectivity failure	Handle failed/incomplete requests safely and ensure transactions do not corrupt data; verify network availability before dependent operations.
Incorrect RBAC permissions	Test each defined role separately and enforce authorization at protected operations.
Patient information exposure	Use HTTPS/TLS, role-based access, secure password hashing, audit logs, and controlled access to medical records.
External service failure	Use defined interfaces and handle payment/SMS/email/laboratory/insurance service failures without corrupting HMS data.
Appointment double booking	Validate doctor availability and apply scheduling logic that rejects an occupied slot.
Data loss or corruption	Use controlled database operations, consistency checks, backups, and recovery procedures.

3.8 Traceability to Requirements

Requirement	Architectural Element
HMS-F-001 to HMS-F-003	Authentication, session management, and RBAC components
HMS-F-004 to HMS-F-006	Patient Management + RBAC + Centralized Database
HMS-F-007 to HMS-F-008	Doctor Management + Appointment/Patient services
HMS-F-009 to HMS-F-012	Appointment Management + Notification component
HMS-F-013 to HMS-F-015	Medical Record Management + RBAC + Centralized Database
HMS-F-016 to HMS-F-017	Prescription Management + Medicine data
HMS-F-018 to HMS-F-019	Billing & Payment Management
HMS-F-020	Reports component + centralized hospital data
HMS-NF-001	Application/service performance design and monitoring
HMS-NF-002	Availability, monitoring, backup/recovery design
HMS-NF-003	Security layer + HTTPS/TLS
HMS-NF-004	Database/data access layer + transaction/data consistency controls
HMS-NF-005	Presentation/UI design and usability controls
HMS-SR-001 to HMS-SR-005	Security layer: RBAC, TLS, password hashing, audit logging, session timeout

3.9 Security Architecture

Security is designed around confidentiality, integrity, and availability requirements stated in the SRS. The security architecture applies controls at authentication, authorization, communication, data, session, and audit levels.

Security Concern	Design Control
Authentication	Username/email and password authentication; secure session creation and termination.
Authorization	RBAC for patient, doctor, receptionist, pharmacist, administrator, and system administrator roles.
Confidentiality	HTTPS with TLS 1.2 or higher; restricted access to sensitive records.
Password Protection	Strong one-way password hashing; passwords are never stored in plain text.
Session Security	Automatic termination or locking after configured inactivity; secure logout.
Auditability	Audit logs for authentication, patient record modifications, billing changes, prescription changes, and administrative actions.
Integrity	Controlled updates, authorization checks, and database consistency mechanisms.
Availability	Monitoring, backups, recovery procedures, and handling of network/database failures.

Threat / Control Mapping

Threat Area	Potential Impact	Primary Control
Spoofing / unauthorized login	Unauthorized account access	Authentication, secure password hashing, session controls
Tampering	Unauthorized record modification	RBAC, authorization checks, audit logging, data integrity controls
Information disclosure	Exposure of patient/medical data	RBAC, HTTPS/TLS, controlled record access
Denial of Service / service failure	Reduced availability	Availability monitoring, failure handling, recovery procedures
Privilege escalation	Access to functions outside assigned role	Centralized RBAC and authorization enforcement

4. Design

4.1 Design Overview

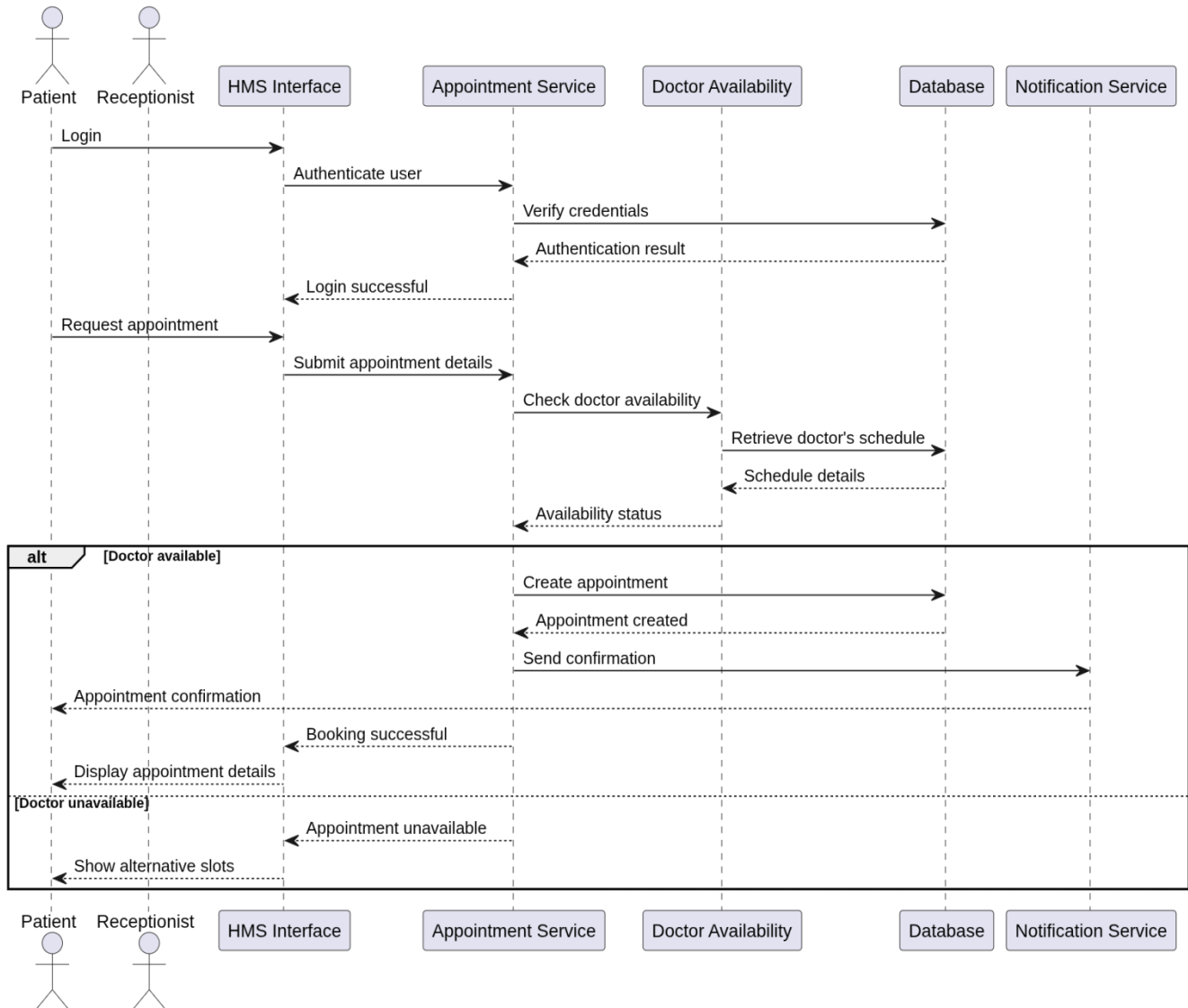
The HMS is designed as a centralized, modular system. User requests enter through a role-specific interface, pass through authentication and authorization controls, are handled by the relevant application/domain module, and are persisted in the centralized database. External services are accessed through controlled integration interfaces. This design supports separation of concerns, security, data consistency, and maintainability.

4.2 UML Sequence Diagrams

At least two sequence diagrams are required. The following flows are directly supported by the SRS and STP.

Sequence Diagram 1 — Appointment Scheduling

HMS - Appointment Scheduling Sequence Diagram

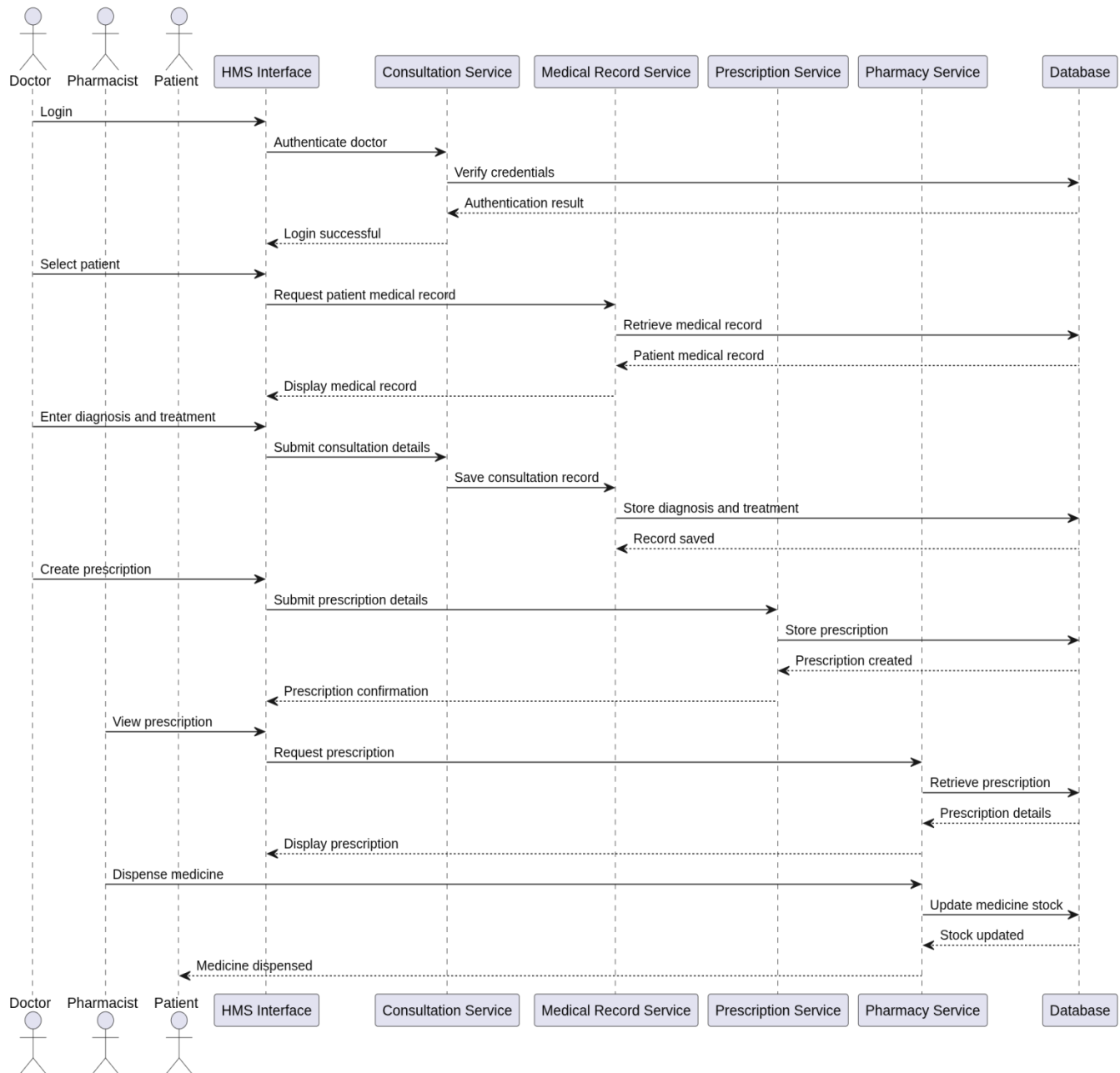


Suggested participants: Patient/Receptionist → HMS UI → Authentication/RBAC → Appointment Service → Doctor Availability → Patient Records → Centralized Database → Notification Service.

1. User authenticates and receives an authorized session.
2. User selects a doctor, date, and time slot.
3. Appointment service checks patient and doctor availability.
4. System rejects the request if the selected slot is already occupied.
5. For an available slot, the appointment is stored in the centralized database.
6. Confirmation/reminder notification is generated through the notification interface.

Sequence Diagram 2 — Doctor Consultation and Prescription

HMS - Doctor Consultation and Prescription Sequence Diagram



Suggested participants: Doctor → HMS UI → Authentication/RBAC → Patient Record Service → Medical Record Database → Prescription Service → Centralized Database.

7. Doctor authenticates and accesses permitted appointments/patient information.
8. System verifies that the doctor is authorized to access the selected patient record.
9. Doctor records diagnosis, treatment details, and consultation notes.
10. Medical record service stores the updated record.
11. Doctor creates an electronic prescription containing medicine, dosage, frequency, duration, and instructions.
12. Prescription service stores the prescription for authorized viewing/dispensing.

4.3 API / Interface Design

The source documents define logical software interfaces but do not prescribe exact REST endpoint names, request schemas, or response schemas. The following interfaces should therefore be treated as design-level contracts to be finalized during implementation.

Interface	Method / Interaction	Purpose	Key Data
Authentication Interface	Request/Response	Authenticate user and establish/terminate session.	Username/email, password, session status
Appointment Interface	Request/Response	Schedule, cancel, reschedule, and validate appointment availability.	Patient, doctor, date, time slot, appointment status
Patient Record Interface	Request/Response	Retrieve and update authorized patient information and medical records.	Patient ID, demographic data, medical record data, authorization context
Prescription Interface	Request/Response	Create and retrieve prescriptions and dispensing information.	Patient, doctor, medicines, dosage, frequency, duration, instructions
Billing Interface	Request/Response	Generate bills and record payments.	Patient, services, charges, total, payment status
Notification Interface	Request/Response	Send appointment confirmations and reminders.	Appointment, recipient, notification message/status
External Payment Interface	API integration	Connect HMS billing/payment functions with an external payment service where applicable.	Payment request/status; exact schema TBD

Example logical error categories: authentication failure, authorization failure, invalid input, unavailable appointment slot, record-not-found, database/service failure, and external-service failure. Exact HTTP status codes and endpoint paths are to be finalized with the implementation API specification.

4.4 Error Handling, Logging & Monitoring

- Validate user input and provide understandable validation messages.
- Reject unauthorized operations and prevent access to protected functions and records.
- Reject appointment requests when the selected doctor/time slot is unavailable.
- Handle database or external-service failures without corrupting HMS records.
- Terminate or lock inactive sessions according to the configured timeout.
- Maintain audit logs for authentication events, patient record modifications, billing changes, prescription changes, and administrative actions.
- Avoid exposing sensitive patient or authentication information in general error messages or logs.
- Monitor availability, response time, failures, and other operational indicators required to verify the SRS quality attributes.

4.5 UX Design

The UI should provide a clear and consistent experience with role-specific dashboards. The design should support the common workflows identified in the SRS and STP, including patient registration, appointment scheduling, medical record access, prescription handling, billing, payment recording, and report generation.

- Role-specific dashboards and permitted functions.
- Clear validation messages for incorrect, incomplete, or invalid data.
- Simple navigation between patient records, appointments, prescriptions, billing, and relevant functions.
- Consistent feedback for successful and failed operations.
- Usable workflows that allow common hospital tasks with minimal training.
- Responsive support for supported desktop, laptop, and tablet environments.

4.6 Open Issues & Next Steps

Open Issue	Next Step
Programming language and application framework not specified in SRS	Select and document implementation stack during development.
Database product not specified	Select the centralized DBMS and document schema, indexing, backup, and recovery design.
Exact API endpoint names and payloads not specified	Prepare detailed API specification before implementation/integration testing.
External service contracts not specified	Define interfaces for payment, SMS/email, laboratory, pharmacy, and insurance integrations where required.
Detailed UML component/sequence diagrams	Create final UML diagrams from the approved implementation design.
Monitoring implementation details	Define metrics, dashboards, alerts, and operational thresholds during deployment design.
Final deployment topology	Document production/test environment configuration, network boundaries, backups, and recovery procedures.

5. Appendices

5.1 Glossary

Term	Meaning
HMS	Hospital Management System
RBAC	Role-Based Access Control
EMR	Electronic Medical Record
API	Application Programming Interface
SRS	Software Requirements Specification
STP	Software Test Plan
RTM	Requirements Traceability Matrix
UAT	User Acceptance Testing
TLS	Transport Layer Security
HTTPS	Secure HTTP communication

5.2 References

- Hospital Management System Software Requirements Specification (SRS), Version 1.0, Team 13, dated 12-09-2026.
- Hospital Management System Software Test Plan (STP).
- Hospital Management System Requirements Traceability Matrix (RTM).
- Hospital Management System UML Use-Case Diagrams.

5.3 Tools

UML diagrams (class and sequence) PlantUML

Component diagram PlantUML

Documentation Microsoft Word

Version control and collaboration GitHub

API documentation and testing Postman / Swagger (OpenAPI)